

ROCK FALLS, IL, USA

WMO: 725326

Lat: 41.743N

Lon: 89.676W

Elev: 197

StdP: 98.98

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04894

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.8	-17.9	-26.4	0.3	-20.9	-22.8	0.5	-17.6	12.7	-5.3	11.6	-6.8	2.8	280	0.515

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	32.6	24.1	31.3	23.5	29.9	22.6	25.7	30.9	24.7	29.6	23.7	28.2	3.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.0	19.4	29.2	22.9	18.1	27.8	22.3	17.4	27.1	80.5	31.0	75.8	29.3	71.9	28.3	32.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.2	9.4	8.3	-25.5	35.3	3.4	1.8	-28.0	36.5	-30.0	37.6	-31.9	38.6	-34.4	39.8		
(5)			-26.3	27.6	3.4	1.5	-28.8	28.7	-30.8	29.5	-32.6	30.4	-35.1	31.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	-5.8	-3.5	3.0	9.9	16.0	21.4	22.9	21.8	18.1	11.4	4.0	-2.6		
	DBStd	11.32	7.04	6.92	5.97	5.05	4.77	3.61	2.99	2.86	4.35	5.12	5.59	6.35		
	HDD10.0	1795	489	379	229	61	5	0	0	0	1	44	193	392		
	HDD18.3	3574	748	612	476	256	105	13	2	5	57	221	431	649		
	CDD10.0	1717	0	1	12	58	191	342	399	366	244	89	12	2		
	CDD18.3	453	0	0	1	3	33	106	142	112	50	7	0	0		
	CDH23.3	4399	0	0	3	59	338	1022	1374	967	541	93	1	0		
	CDH26.7	1446	0	0	0	9	99	358	479	298	181	22	0	0		
(14) Wind	WSAvg	3.5	4.1	4.1	4.2	4.5	3.9	3.0	2.5	2.3	2.7	3.4	3.8	3.8		
(15) Precipitation	PrecAvg	939	44	41	62	91	115	124	98	109	75	68	69	45		
	PrecMax	1228	76	96	133	153	255	286	254	190	180	205	135	90		
	PrecMin	604	11	8	18	41	16	31	9	32	20	10	7	10		
	PrecStd	155	19	24	40	36	60	73	52	42	43	49	36	22		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.7	16.6	22.8	28.0	32.2	33.7	34.6	33.6	33.0	29.0	22.1	16.1		
		MCWB	10.3	12.4	15.6	17.0	22.0	23.7	26.2	24.6	23.6	21.0	15.2	13.6		
	2%	DB	7.7	12.0	18.1	25.5	29.1	32.3	32.4	31.4	31.1	26.2	17.7	12.0		
		MCWB	5.7	9.0	13.0	16.0	19.7	23.3	25.2	24.2	22.4	17.8	12.6	9.7		
	5%	DB	5.1	7.9	15.2	22.4	27.3	30.9	31.1	29.7	28.7	22.8	15.3	7.8		
		MCWB	3.5	5.3	11.0	14.2	18.7	22.6	24.5	23.3	21.0	16.4	11.6	6.0		
	10%	DB	2.7	5.1	12.2	19.1	24.9	28.7	29.1	27.9	27.0	20.6	12.7	5.8		
		MCWB	1.3	2.9	8.2	12.6	17.5	21.5	23.2	21.8	19.4	15.4	9.2	3.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.9	12.8	16.7	19.1	23.1	26.1	27.7	26.1	25.1	22.1	16.1	14.1		
		MCDB	12.1	14.7	21.8	25.3	29.6	31.5	32.8	31.9	31.1	27.8	19.5	16.3		
	2%	WB	6.2	8.7	14.0	16.9	21.5	24.6	26.2	25.1	23.6	19.6	14.0	9.7		
		MCDB	7.7	11.2	17.5	23.0	27.5	29.9	31.2	30.3	28.5	23.3	17.2	11.6		
	5%	WB	3.5	5.6	11.0	15.3	20.0	23.5	25.2	24.0	22.1	17.6	11.8	6.4		
		MCDB	5.1	8.0	15.2	21.1	25.3	28.7	29.9	28.4	27.0	22.0	14.7	8.0		
	10%	WB	1.6	3.1	8.2	13.5	18.7	22.3	24.0	22.9	20.6	15.6	9.7	3.6		
		MCDB	2.8	5.1	11.7	18.9	23.5	27.4	28.1	26.9	25.2	19.7	12.9	5.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.4	9.0	10.2	12.1	11.6	11.5	11.4	11.3	13.2	12.0	9.7	8.0		
		MCDBR	9.8	11.7	14.6	16.7	14.0	13.2	12.4	12.9	15.1	15.6	13.6	10.3		
	5% WB	MCWBR	8.4	9.3	10.0	9.9	7.8	6.7	6.7	7.1	7.5	9.2	9.9	8.8		
		MCWBR	9.4	11.5	13.7	14.4	12.2	11.5	11.3	11.2	12.6	13.4	11.9	9.7		
(40) Clear-Sky Solar Irradiance	taub	taub	0.289	0.304	0.341	0.388	0.421	0.425	0.447	0.425	0.397	0.344	0.303	0.286		
		taud	2.410	2.379	2.377	2.257	2.227	2.262	2.188	2.247	2.297	2.446	2.501	2.439		
	Ebn at Noon	Ebn at Noon	864	904	905	881	857	852	828	835	833	847	843	836		
		Edh at Noon	79	95	108	131	139	134	143	131	116	89	72	71		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.74	2.70	3.70	4.66	5.44	6.14	6.09	5.38	4.52	2.98	1.95	1.43		
	RadStd	RadStd	0.20	0.32	0.33	0.40	0.37	0.49	0.37	0.33	0.39	0.35	0.16	0.16		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page